Exam Date & Time: 27-Sep-2024 (10:45 AM - 12:15 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech III Semester Sessional Examination September 2024

ELECTROMAGNETIC WAVES [ECE 2125]

Duration: 90 mins.

Marks: 30

MCQ

Section Duration: 20 mins

Answer all the questions.

1) Which of these is not valid at point (0, 4, 0)? (1)

1) $a_\varphi = -a_x$ 2) $a_\theta = -a_z$ 3) $a_r = 4a_y$ 4) $a_\rho = a_y$

2) If infinite sheet of charges of +2C/m^2 and -2 C/m^2 are located at z = +2 and z = -3 respectively, then the flux density at the origin is _____ (1)

1) 0 C/m^2 2) +2 $\mathbf{a_z}$ C/m^2 3) +4 $\mathbf{a_z}$ C/m^2 4) -2$\mathbf{a_z}$ C/m^2

3) If Electric scalar potential is given by V = 3x +5, the magnitude of Electric field at P(2,3,5) is___ (1)

1) 3 2) 11 3) 8 4) 6

4) Find H = _____ A/m at the center of a circular coil of diameter 1 m and carrying a current of 2 A. (1)

1) 6.366 2) 2 3) 0.6366 4) 0.2

5) Calculate the flux density at a distance of 5 cm from a long straight circular conductor carrying a current of 250 A and placed in air. (1)

1) 10^{-2} T 2) 10^3 T 3) 10^2 T 4) 10^{-3} T

DESCRIPTIVE

Answer all the questions.

6) Convert the vector to cartesian coordinates: $\boldsymbol{D} = \frac{\sin\theta}{r^2}\boldsymbol{a_r} + \frac{\cos\theta}{r^2}\boldsymbol{a_\theta}$. (2)

7) (3)

Two uniform vector fields are given by $\boldsymbol{E} = -5\boldsymbol{a}_\rho + 10\boldsymbol{a}_\varphi + 3\boldsymbol{a}_z$ and $\boldsymbol{F} = \boldsymbol{a}_\rho + 2\boldsymbol{a}_\varphi - 6\boldsymbol{a}_z$. Calculate

a. $|\boldsymbol{E} \boldsymbol{X} \boldsymbol{F}|$

b. The vector component of **E** at P(5, π/2, 3) parallel to the line x = 2, z = 3

c. The angle **E** makes with the surface z = 3 at P.

13

8) If $V = 60\sin\theta/r^2$ V in free space and point P is located at ($r = 3$, $\theta = 60^0$, $\varphi = 25^0$) Find

a. V_P

b. E_P

c. a_N

d. D

e. ρ_V

(3)

9) Determine the value of divergence D at the point P (0.3,0.4,0.5) if (3)

$\mathbf{D} = 20xy^2(z+1)\,\mathbf{ax} + 20x^2\,y(z+1)\mathbf{a_y} + 10\,x^2\,y^2\,\mathbf{a_z}$ C/m^2 .

10) Starting from the expression of electric field of a point charge, derive the expression of absolute potential due to point charge. (2)

11) Evaluate the closed line integral of H from P1(5, 4, 1) to P2(5, 6, 1) to P3(0, 6, 1) to P4(0, 4, 1) to P1, using straight line segments, If $H = 0.1y^3\,\mathbf{a_x} + 0.4x\,\mathbf{a_z}$ A/m. (3)

12) In a constant magnetic field of 0.6 Tesla along the z-direction, find the value of the path (2)

integral $\oint A \cdot dl$ in the units of (Tesla m^2) on a square loop of side length $\frac{1}{\sqrt{2}}$ metres. The normal to the loop makes an angle of 60° to the z-axis, as shown in the Figure 12.

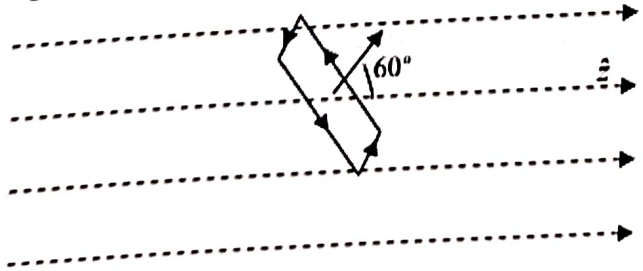


Figure 12.

3) Two parallel sheet separated by a distance 'k' from each other is kept along z-axis. Find the magnetic field intensity vector H for the region $z < 0$ (considered as region 1), $0 < z < k$ (considered as region 2) and $z > k$ (considered as region 3) if both the sheets carry surface current $K = K_y a_y$. (3)

14) Two point charges Q1 and Q2 are located at (1, 2, 0) m and (2, 0, 0) m respectively. Find the relation between the charges Q1 and Q2 such that the total force on a test charge at P(-1, 1, 0) m will have (4)

a. no x-component

b. no y component.

-----End-----

EMW – Midterm 2024

Scheme of valuation

6. Two uniform vector fields are given by $\boldsymbol{E} = -5\boldsymbol{a}_\rho + 10\boldsymbol{a}_\varphi + 3\mathbf{a}_z$ and $\boldsymbol{F} = \boldsymbol{a}_\rho + 2\boldsymbol{a}_\varphi - 6\boldsymbol{a}_z$. Calculate (a) $|\boldsymbol{E}\ X\ \boldsymbol{F}|$ (b) The vector component of **E** at P(5, π/2, 3) parallel to the line x = 2, z = 3 (c) The angle **E** makes with the surface z = 3 at P.
(3M)

Solution:
A) 74.06
B) $-5\boldsymbol{a}_\rho\ (OR\ -5\boldsymbol{a}_y)$
C) 15.02 degrees

7. Convert the vector to cartesian coordinates: $\boldsymbol{D} = \frac{\sin\theta}{r^2}\boldsymbol{a}_r + \frac{\cos\theta}{r^2}\boldsymbol{a}_\theta$ (2M)

$$\begin{bmatrix} D_x \\ D_y \\ D_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\phi & 0 \end{bmatrix} \begin{bmatrix} \frac{\sin\theta}{r^2} \\ \frac{\cos\theta}{r^2} \\ 0 \end{bmatrix}$$

$$D_x = \frac{\sin^2\theta\cos\phi}{r^2} + \frac{\cos^2\theta\cos\phi}{r^2} = \frac{\cos\phi}{r^2} = \frac{x}{\sqrt{x^2+y^2}(x^2+y^2+z^2)};$$

$$D_y = \frac{\sin^2\theta\sin\phi}{r^2} + \frac{\cos^2\theta\sin\phi}{r^2} = \frac{\sin\phi}{r^2} = \frac{y}{\sqrt{x^2+y^2}(x^2+y^2+z^2)};$$

$$D_z = \frac{\sin\theta\cos\theta}{r^2} - \frac{\sin\theta\cos\theta}{r^2} = 0;$$

$$\therefore\quad D = \frac{I}{\sqrt{x^2+y^2}(x^2+y^2+z^2)}(x\bar{a}_x + y\bar{a}_y)$$

8.8. Given $V = \frac{60\sin\theta}{r^2}$, $P(r=3, \theta=60°, \phi=25°)$

(a) $V_P = \frac{60\sin 60°}{(3)^2} = 5.774\ V$

(b) $\bar{E}_P = -\nabla V = -\left[\frac{\partial V}{\partial r}\bar{a}_r - \frac{1}{r}\frac{\partial V}{\partial \theta}\bar{a}_\theta\right]$ — 1M

$\bar{E}_P = 3.849\,\bar{a}_r - 1.11\,\bar{a}_\theta\ V/m$

(c) $\bar{a}_N = \frac{\bar{E}_P}{|\bar{E}_P|} = -0.96\,\bar{a}_r + 0.277\,\bar{a}_\phi$ — 1M

(d) $\bar{D} = \epsilon\bar{E} = \epsilon_0\left[\frac{120\sin\theta}{r^3}\bar{a}_r - \frac{60\cos\theta}{r^3}\bar{a}_\theta\right]$ — 1M

(e) $\rho_v = \nabla\cdot\bar{D} = -\frac{120\epsilon_0\sin\theta}{r^4} - \frac{60\epsilon_0\cos 2\theta}{\sin\theta}$ — 1M

$\rho_{v(P)} = -7.57\ pC/m^3$

9. Given $\bar{D} = 20xy^2(z+1)\bar{a}_x + 20x^2y(z+1)\bar{a}_y$

$+ 10x^2y^2\bar{a}_z$ c/m²

@(0.3, 0.4, 0.5)

div $\bar{D}$ = ?

$$\text{div}\bar{D} = \frac{\partial}{\partial x}(D_x) + \frac{\partial}{\partial y}(D_y) + \frac{\partial}{\partial z}(D_z)$$ -1M

$$\text{div}\bar{D} = 20y^2(z+1) + 20x^2(z+1) + 0$$ -2M

$$\text{div}\bar{D}_P = 7.5 \text{ c/m}^3$$

Q10) Electric field due to point charge $= \vec{E} = \frac{Q}{4\pi\epsilon r^2}\hat{a}_r$ placed in origin — 0.5

$$V_r = -\int_{\infty}^{r} \vec{E}\cdot d\vec{l} = \frac{Q}{4\pi\epsilon}\left[\frac{1}{r}\right]_{\infty}^{r} \quad \text{— 0.5}$$

$$V_r = \frac{Q}{4\pi\epsilon r} \quad \text{— 0.5}$$

Q11) $\int_{P_1}^{P_2} \vec{H}\cdot d\vec{l} = \int_{1}^{2} \vec{H}\cdot dy\hat{a}_y = \int_{4}^{6} 0 = 0$ — 0.5

$$\int_{P_2}^{P_3} \vec{H}\cdot d\vec{l} = \int_{2}^{3} \vec{H}\cdot dx\hat{a}_x = \int_{5}^{0} 0.1y^3 dx\Big|_{y=6} = 0.1(6^3)[x]_5^0$$

$$= -108 \quad \text{— 0.5}$$

$$\int_{P_3}^{P_4} \vec{H}\cdot d\vec{l} = \int_{3}^{4} \vec{H}\cdot dy\hat{a}_y = \int_{6}^{4} 0 = 0 \quad \text{— 0.5}$$

$$\int_{P_4}^{P_1} \vec{H}\cdot d\vec{l} = \int_{4}^{1} \vec{H}\cdot dx\hat{a}_x = \int_{0}^{5} 0.1y^3 dx\Big|_{y=4} = 0.1(4)^3[x]_0^5$$

$$= 32 \quad \text{— 0.5}$$

$$\oint \vec{H}\cdot d\vec{l} = 0 - 108 + 0 + 32 = -76\,A \quad \text{— 1}$$

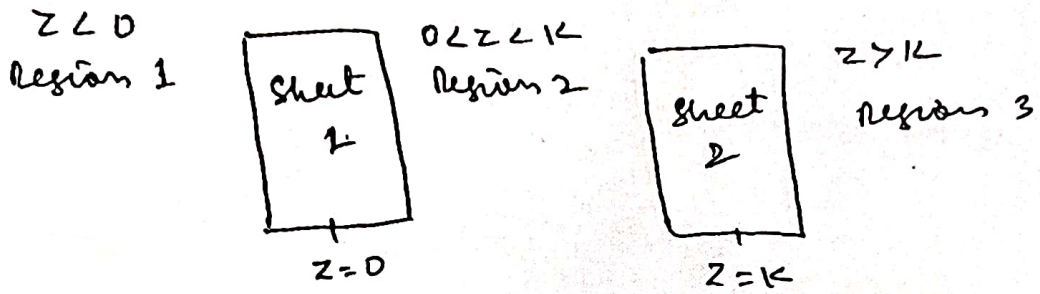
Q12:- $\oint \vec{A} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{A}) \cdot \vec{da} = \int_S \vec{B} \cdot \vec{da} = B A \cos 60^\circ$

$\therefore$ $B = 0.6$ Tesla, Area (A) of square $= \left(\frac{1}{\sqrt{2}}\right)^2 m^2$

On putting the above value

$$\oint \vec{A} \cdot d\vec{l} = 0.6 \times \left(\frac{1}{\sqrt{2}}\right)^2 \times \frac{1}{2} = 0.15 \text{ Tesla } m^2$$

Q13



For Region 1 ($z < 0$)

$$H = H_{sheet\,1} + H_{sheet\,2} = \frac{K_y(a_y x - a_z)}{2} + \frac{K_y(a_y x - a_z)}{2}$$

$$= -K_y a_x$$

For Region 2 ($0 < z < K$)

$$H = H_{sheet\,1} + H_{sheet\,2} = \frac{K_y(a_y \times a_z)}{2} + \frac{K_y(a_y x - a_z)}{2}$$

$$= 0$$

For Region 3 ($z > K$)

$$H = H_{sheet\,1} + H_{sheet\,2} = \frac{K_y(a_y \times a_z)}{2} + \frac{K_y(a_y \times a_z)}{2} = K_y a_x$$

14) $E = E_1 + E_2 = \frac{Q_1}{4\pi\epsilon r_1^2}\hat{a}_{r1} + \frac{Q_2}{4\pi\epsilon r_2^2}\hat{a}_{r2}$ (V/m)

$r_1 = -2\hat{a}_x - \hat{a}_y$, $|r_1| = \sqrt{5}$, $\hat{a}_{r1} = \frac{-2}{\sqrt{5}}a_x - \frac{1}{\sqrt{5}}\hat{a}_y$ (1M)

$r_2 = -3a_x + a_y$, $|r_2| = \sqrt{10}$, $\hat{a}_{r2} = \frac{-3}{\sqrt{10}}\hat{a}_x + \frac{1}{\sqrt{10}}\hat{a}_y$ (1M)

$$E = \frac{Q_1}{4\pi\epsilon\times 5}\left(\frac{-2}{\sqrt{5}}\hat{a}_x - \frac{1}{\sqrt{5}}\hat{a}_y\right) + \frac{Q_2}{4\pi\epsilon\times 10}\left(\frac{-3}{\sqrt{10}}\hat{a}_x + \frac{1}{\sqrt{10}}\hat{a}_y\right)$$

a) No x-component $E_x = 0$

$$-\frac{2Q_1}{20\pi\sqrt{5}} - \frac{3Q_2}{40\pi\sqrt{10}} = 0 \Rightarrow \frac{Q_1}{Q_2} = \underline{\underline{\frac{-3}{4\sqrt{2}}}}$$ (1M)

b) No y-component $E_y = 0$

$$-\frac{Q_1}{20\pi\sqrt{5}} + \frac{Q_2}{40\pi\sqrt{10}} = 0 \Rightarrow \frac{Q_1}{Q_2} = \underline{\underline{\frac{1}{2\sqrt{2}}}}$$ (1M)